

Crop Yield & Fertiliser Recommendation
BACHELOR OF TECHNOLOGY

in

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by

P.Sushmitha(2520030073)
K.Bindusree(2520030434)

Under the Esteemed guidance of

Dr.Raghavendra
Professor in Dept in CSE



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF CSE

Name of Title: Crop Yield & Fertiliser Recommendation

Abstract:

The Crop Yield & Fertiliser Recommendation System is a machine learning-based project that helps farmers choose a suitable crop and fertiliser based on soil and environmental conditions. The system uses inputs such as Nitrogen (N), Phosphorus (P), Potassium (K), soil pH, rainfall, and temperature to provide suitable recommendations. Machine learning algorithms are trained using agricultural datasets to identify patterns between soil conditions and crop requirements. The main aim of this project is to support better farming decisions, improve crop productivity, and reduce the unnecessary use of fertilisers.

INTRODUCTION:

Agriculture depends greatly on the quality of soil and environmental conditions. Choosing the right crop and using the correct amount of fertiliser can be difficult because different crops require different nutrients and climatic conditions.

The proposed system uses Machine Learning to analyse soil and weather parameters and recommend a suitable crop and fertiliser. The user provides the required values, and the trained model processes them to generate a recommendation.

This project provides a simple and user-friendly approach to agricultural decision-making. It can help reduce guesswork, improve the use of available soil nutrients, and support farmers in achieving better crop growth.

SOFTWARE REQUIREMENTS:

Programming Language

- Python 3.x

Libraries / Frameworks

- **Pandas** – Data handling and preprocessing
- **NumPy** – Numerical calculations
- **Scikit-learn** – Machine learning algorithms and model training
- **Matplotlib** – Data visualization
- **Seaborn** – Graphs and data analysis
- **Streamlit** – Creating a simple web interface



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Development Environment

- Jupyter Notebook / Google Colab
- VS Code (optional)

Operating System

- Windows / Linux / macOS

Dataset

- Crop Recommendation Dataset containing **N, P, K, pH, rainfall, temperature** and crop information

Signature of the Guide

Course Instructor